Communication Systems Lecture-12: Delta Modulation and PTM

Chadi Abou-Rjeily

Department of Electrical and Computer Engineering
Lebanese American University
chadi.abourjeily@lau.edu.lb

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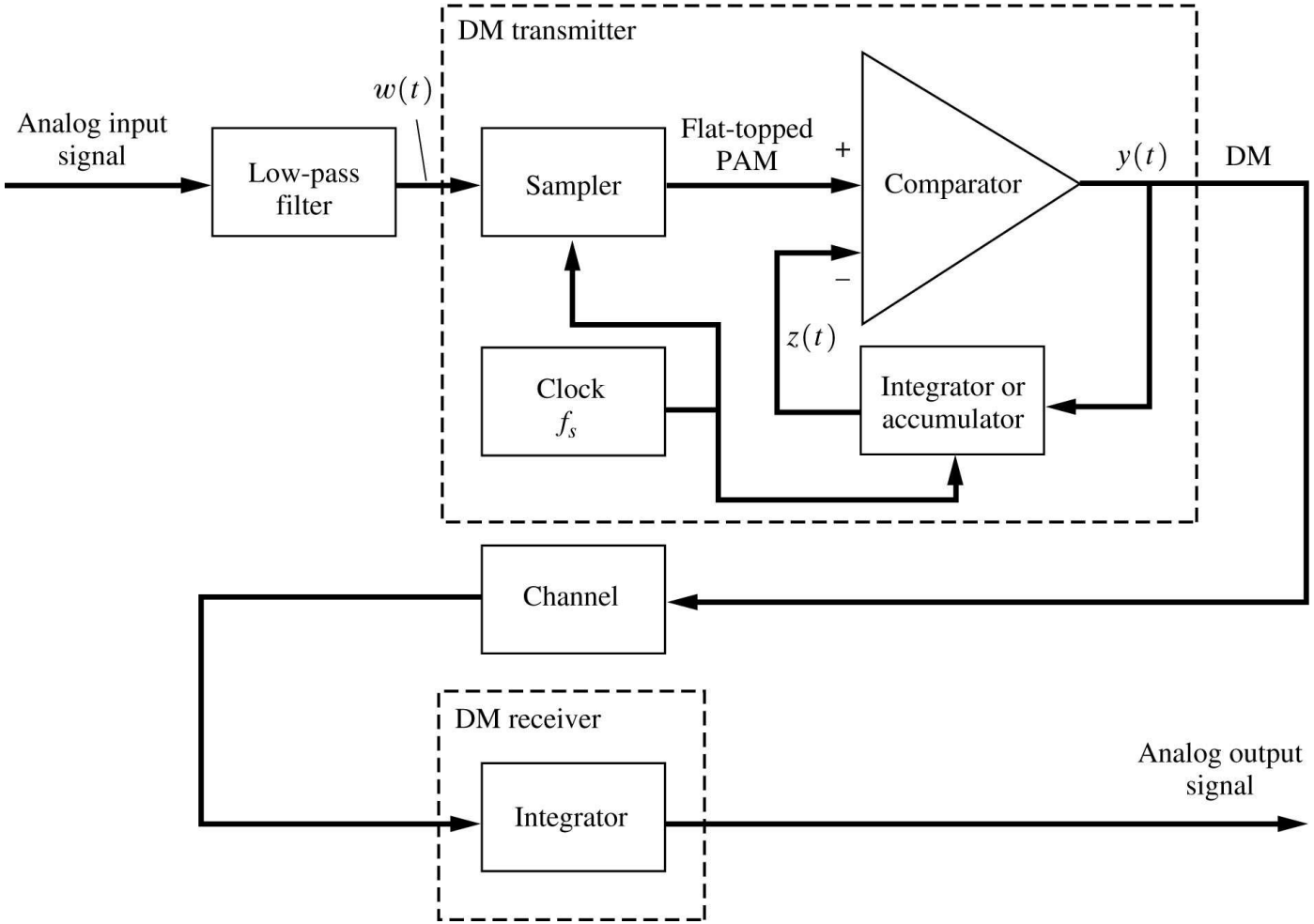
Delta Modulation (1)

- Audio and video information sources are redundant: adjacent samples are usually close to the same value.
- Consequently, the bandwidth and dynamic range of a PCM system are wasted when redundant sample values are retransmitted.
- Bandwidth can be reduced by transmitting PCM signals that correspond to the difference between adjacent samples.
- This can be seen as transmitting the derivative of the signal. At the receiver, integration-like operations are performed to reconstitute the original signal.
- One type of modulation that can be used in this case is the Delta Modulation. This solution is appealing since it is relatively inexpensive.

Delta Modulation (2)

- The next figure shows a DM system:



Delta Modulation (3)

- The output of the comparator can be equal to $\pm V_c$ and the DM signal is a polar signal.
- The integrator's output at time $t = nT_s$ is given by:

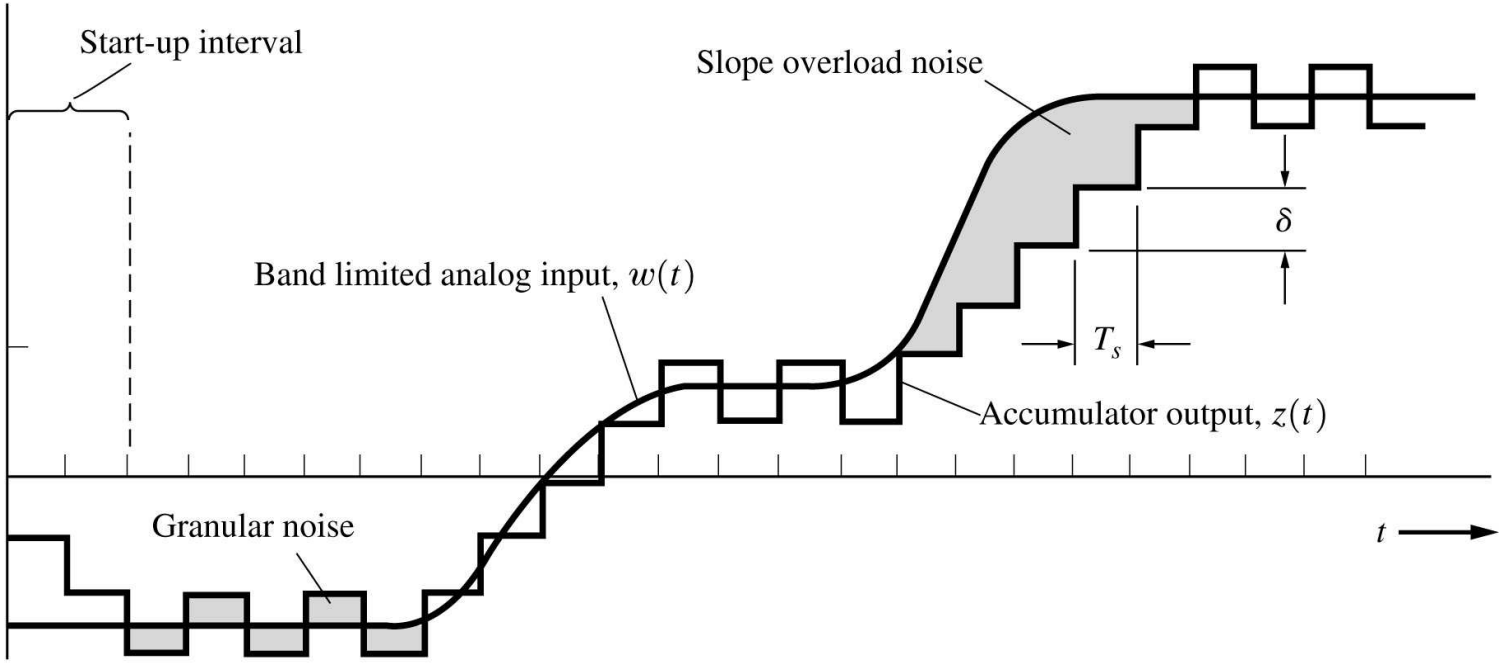
$$z_n = \frac{1}{V_c} \sum_{i=0}^{n} \delta y_i$$

where:

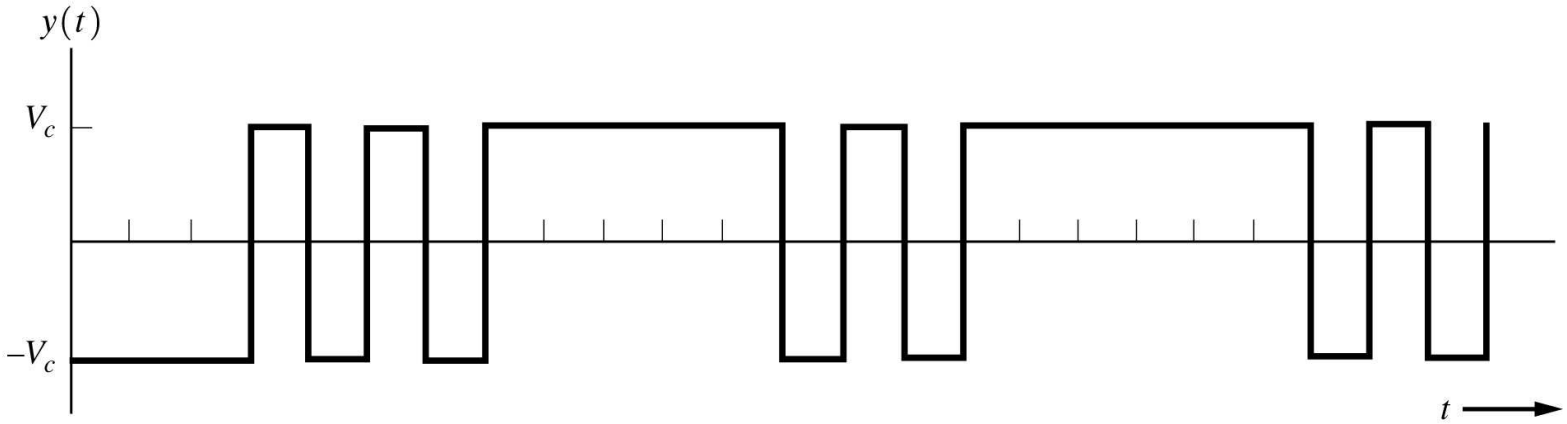
  - $y_i = y(iT_s)$.
  - δ is the accumulator gain or step size.

- At the receiver, demodulation is performed by the use of an integrator. This integrator produces a smoothed version of the accumulator output waveform (that is present at the transmitter).

Delta Modulation (4)



(a) Analog Input and Accumulator Output Waveforms



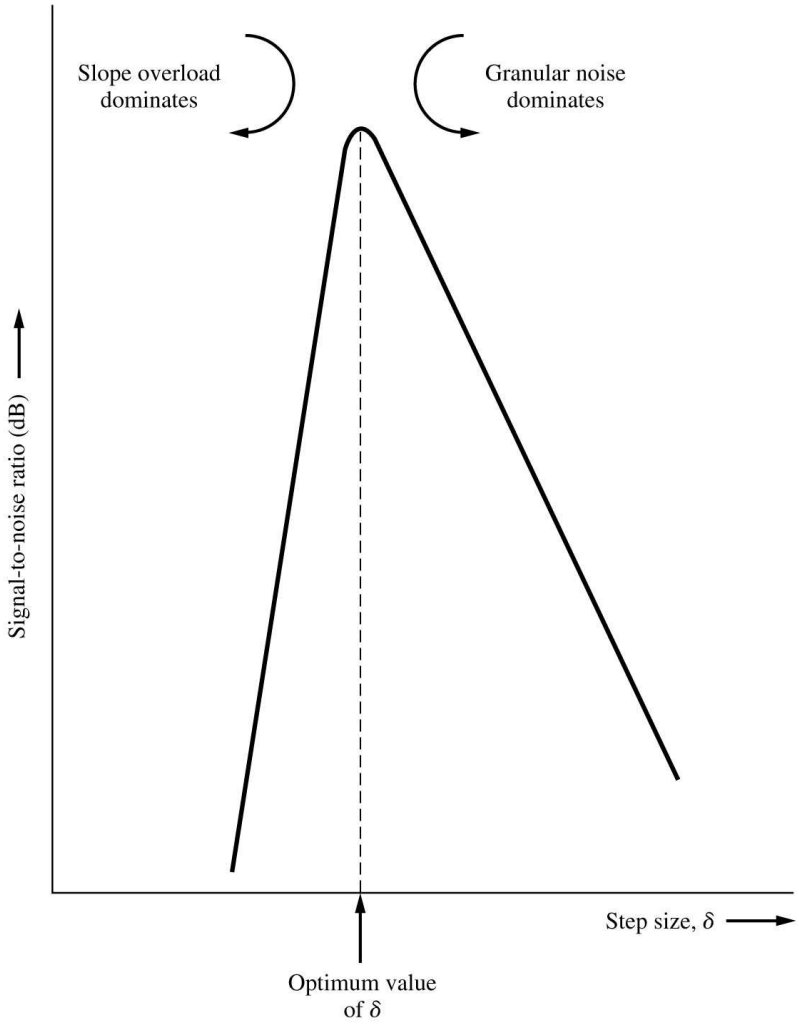
(b) Delta Modulation Waveform

Delta Modulation (5)

- From the previous figure, it can be seen that the accumulator output signal does not always track the analog input signal.
- The quantizing noise error may be classified into two types:
  - Slope overload noise: occurs when the step size δ is too small for the accumulator output to follow quick changes in the input signal.
  - Granular noise: occurs when there is a small variation in the input signal. In this case, the error values are not equally likely from sample to sample. The granular noise decreases for small values of δ.

Delta Modulation (6)

- If δ increases, the granular noise increases while the slope overload noise decreases and vice versa. Consequently, there is an optimum value for δ.



Delta Modulation (7)

Example: Design of a DM system.

- Consider a sine wave input signal: $w(t) = A\sin(\omega_a t)$.
- The slope of $w(t)$ is:

$$\frac{\mathrm{d}w(t)}{\mathrm{d}t} = A\omega_a \cos(\omega_a t)$$

  and the maximum value of the slope is $A\omega_a$.
- On the other hand, the maximum slope that can be generated by the accumulator output is:

$$\frac{\delta}{T_s} = \delta f_s$$

- Consequently, slope overload can be avoided when:

$$\delta f_s > A\omega_a \quad \Rightarrow \quad \delta > \frac{A\omega_a}{f_s} = \frac{2\pi f_a A}{f_s}$$

- On the other hand, δ must not be much larger than this value in order to limit the effect of granular noise.

Delta Modulation (8)

- It can be shown that the PSD for granular noise is $\mathcal{P}_n(f) = \frac{\delta^2}{6f_s}$.
- Consequently, the granular noise power in the analog signal band $|f| < B$ is:

$$N = \langle n^2 \rangle = \int_{-B}^{B} \mathcal{P}_n(f)\mathrm{d}f = \frac{\delta^2 B}{3f_s} = \frac{4\pi^2 A^2 f_a^2 B}{3f_s^3}$$

- On the other hand, the signal power is:

$$S = \langle w^2(t) \rangle = \frac{A^2}{2}$$

- The resulting average signal-to-quantizing-noise ratio is:

$$\left(\frac{S}{N}\right) = \frac{3}{8\pi^2} \frac{f_s^3}{f_a^2 B}$$

  - f_s is the sampling rate, f_a is the frequency of the sinusoidal input and B is the bandwidth of the receiver.

Delta Modulation (9)

- For voice-frequency (VF) audio signals, it was found that the relation $\delta > \frac{2\pi f_a A}{f_s}$ is too restrictive ($f_a = 4$ kHz). In this case, the slope overload is negligible when:

$$\delta > \frac{2\pi 800 W_p}{f_s}$$

  where $W_p = \max|w(t)|$ is the peak value of the audio signal.
- The last equation follows from the fact that the midrange frequencies around 800 Hz dominate in VF signals.
- Consequently, the SNR of DM systems with VF signals is:

$$\left(\frac{S}{N}\right) = \frac{\langle w^2(t)\rangle}{N} = \langle w^2(t)\rangle \frac{3f_s}{\delta^2 B} = \frac{3f_s^3}{(1600\pi)^2 B}\left(\frac{\langle w^2(t)\rangle}{W_p^2}\right)$$

  - B is the audio bandwidth.
  - $\langle w^2(t)\rangle / W_p^2$ is the average-to-peak power ratio.

Delta Modulation (10)

- Assume in what follows that $B = 4$ kHz and that the average-to-peak power ratio is equal to $1/2$.
- In this case, to achieve a SNR of 30 dB, a sampling frequency of $f_s = 40.7$ kHz is required. In other words: $f_s = 10.2B$.
  - Since the output DM signal is a binary signal, this will correspond to a data rate of $R = 1 \times f_s = f_s = 10.2B$.
- On the other hand, the number of bits (n) required to achieve a rate of R with PCM systems is $R = (2B)n$.
- Consequently, the number of bits with PCM needed to achieve the rate of $10.2B$ is $n \approx 5$.
- Given that the average SNR with PCM is 2^{2n}, then the corresponding PCM system will result in a SNR of 30.1 dB.
- In this example, it can be seen that PCM and DM systems having comparable bandwidths result in the same SNR performance.

Delta Modulation (11)

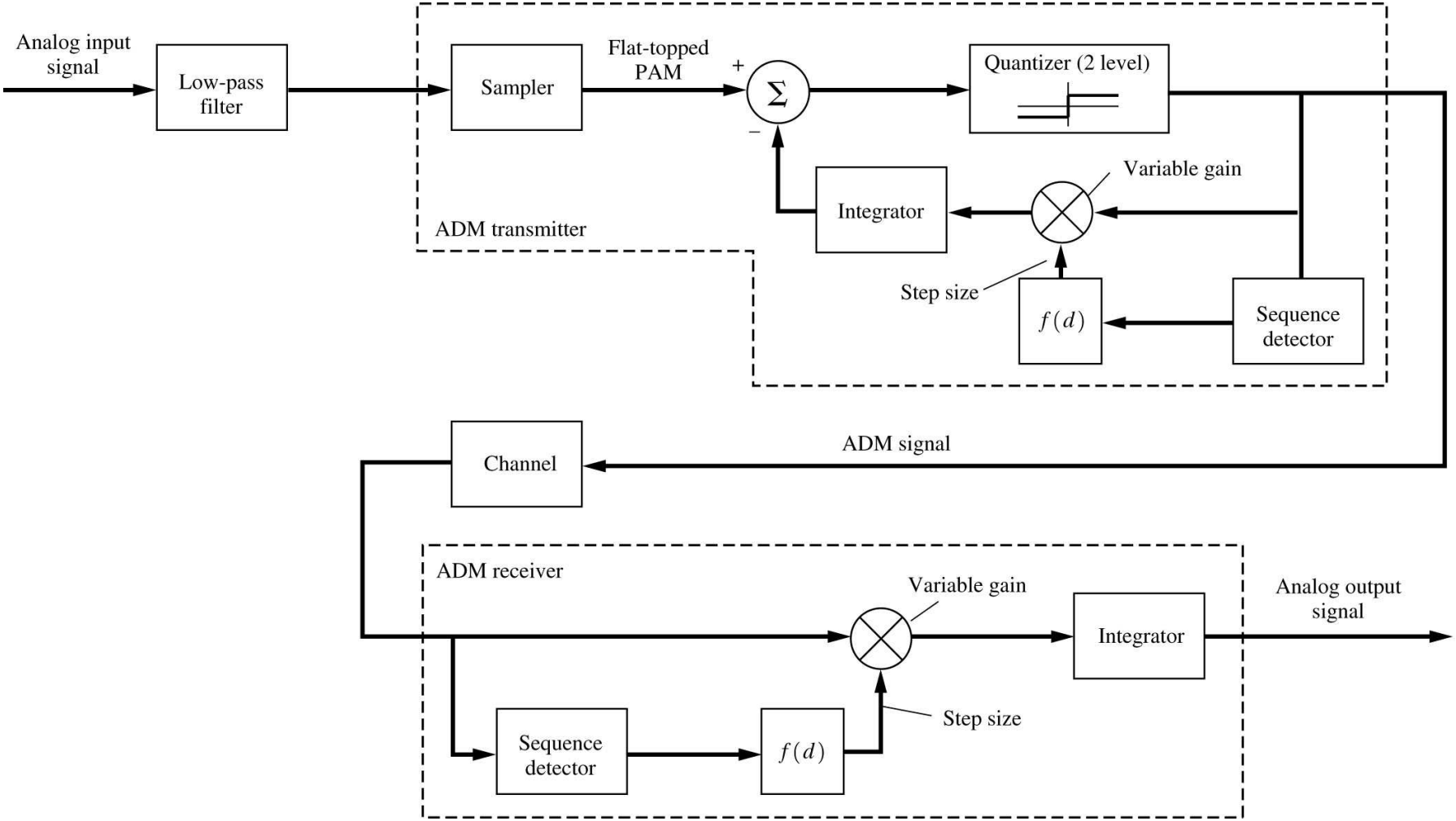
- **In general, for SNRs exceeding 30 dB, PCM has a higher SNR level. For SNRs below 30 dB, DM shows better performance. Note that the comparison between PCM and DM is made with comparable bandwidths.**
- The table below compares the SNRs of DM and PCM when the SNR of DM takes the values of 20 dB, 30 dB and 40 dB:

Modulation Type	Quantity	Unit	Value-1	Value-2	Value-3
DM	SNR	dB	20	30	40
DM	f_s	ksamples/s	18.9	40.7	87.7
DM	$R = 1 \times f_s$	kbits/s	18.9	40.7	87.7
PCM	R	kbits/s	18.9	40.7	87.7
PCM	$f_s = 2B$	ksamples/s	8	8	8
PCM	$n = R/f_s$	unit-less	3	5	11
PCM	SNR	dB	18	30	66

Adaptive Delta Modulation (1)

- To minimize the slope overload noise while holding the granular noise at a reasonable value, adaptive delta modulation (ADM) is used.
- In this case, the step size is varied as the input waveform changes.
- δ takes a sufficiently small value when the input signal does not vary rapidly. This reduces the effect of granular noise.
- δ is increased when the input signal changes rapidly to reduce the slope overload noise.
- δ can be updated by examining the number of consecutive 0's or 1's (since this will indicate that the signal is strictly decreasing or increasing).
- One possible algorithm is to multiply the step size δ by n where n is the number of consecutive 0's or 1's.

Adaptive Delta Modulation (2)

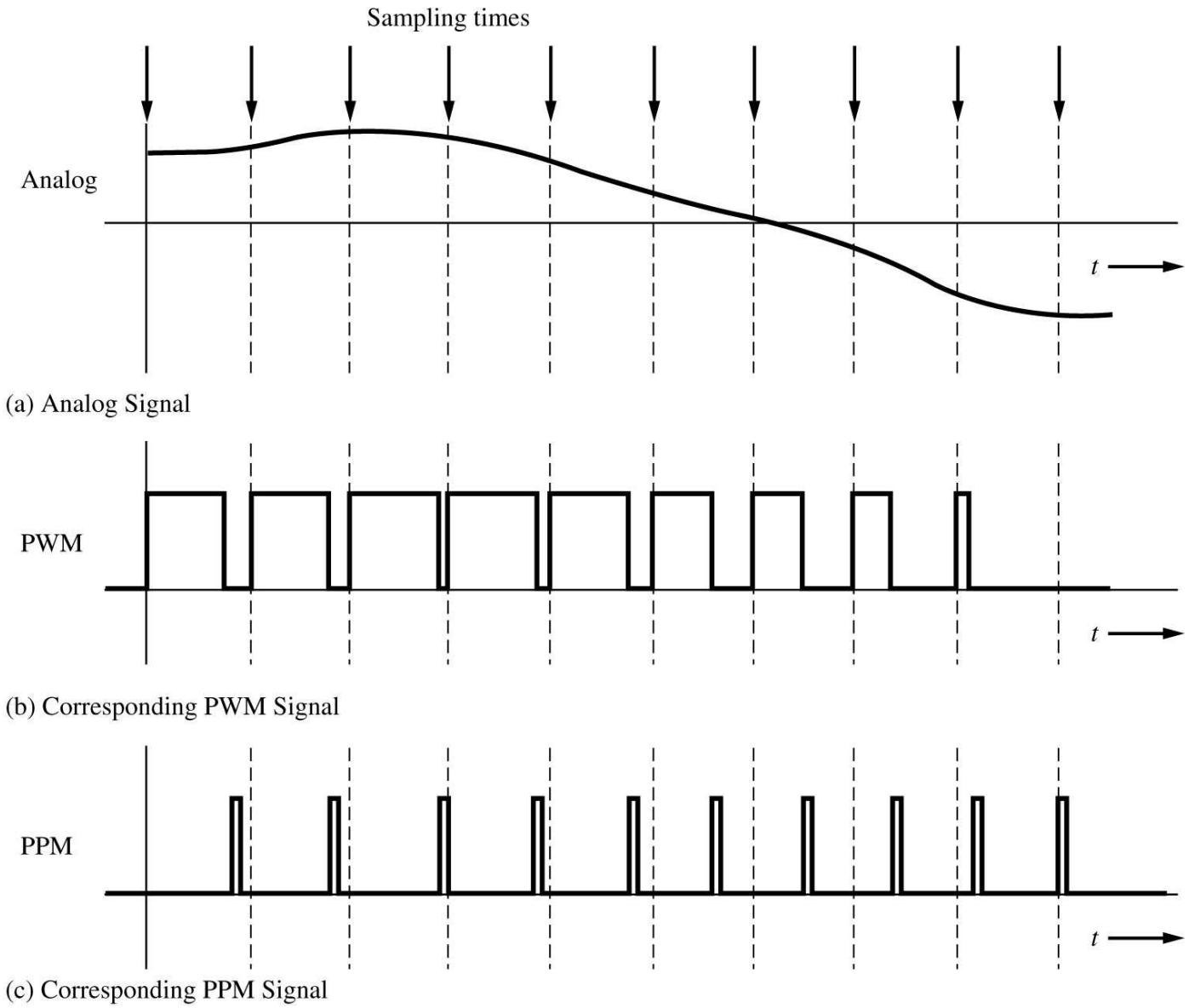


Pulse Time Modulation (1)

Pulse Time Modulation (PTM) encodes the sample values of an analog signal into the time axis of a digital signal. The two main types of PTM are:

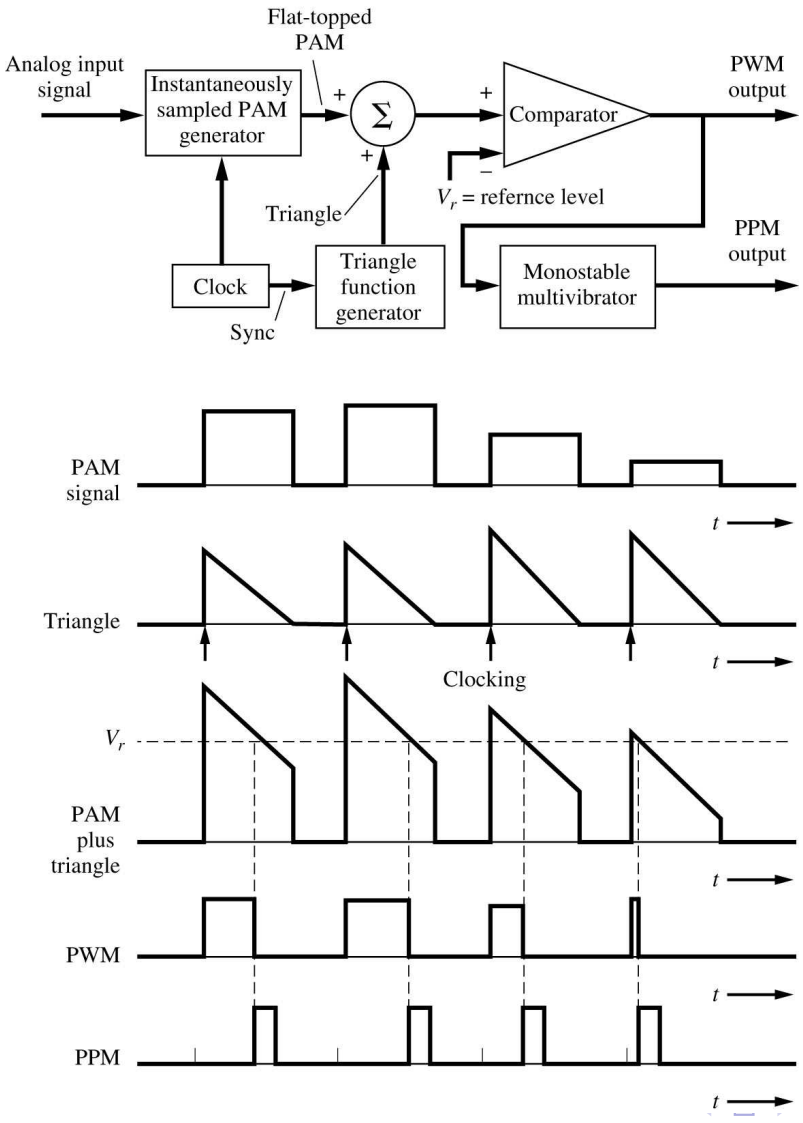
- Pulse Width Modulation (PWM): Sample values of the analog signal determine the width of the transmitted pulse.
  - Either instantaneous or natural sampling can be used.
- Pulse Position Modulation (PPM): Sample values of the analog signal determine the position of a narrow pulse relative to the clocking time.

Pulse Time Modulation (2)



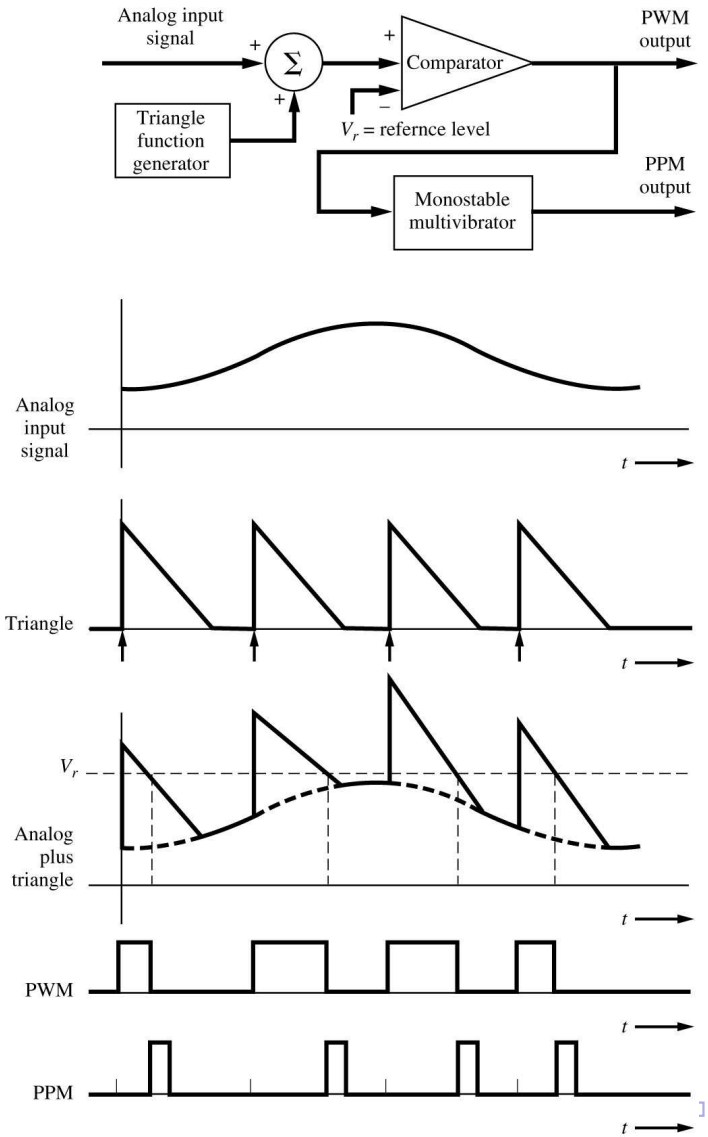
Pulse Time Modulation (3)

- The next figure shows the technique for generating instantaneously sampled PTM signals:



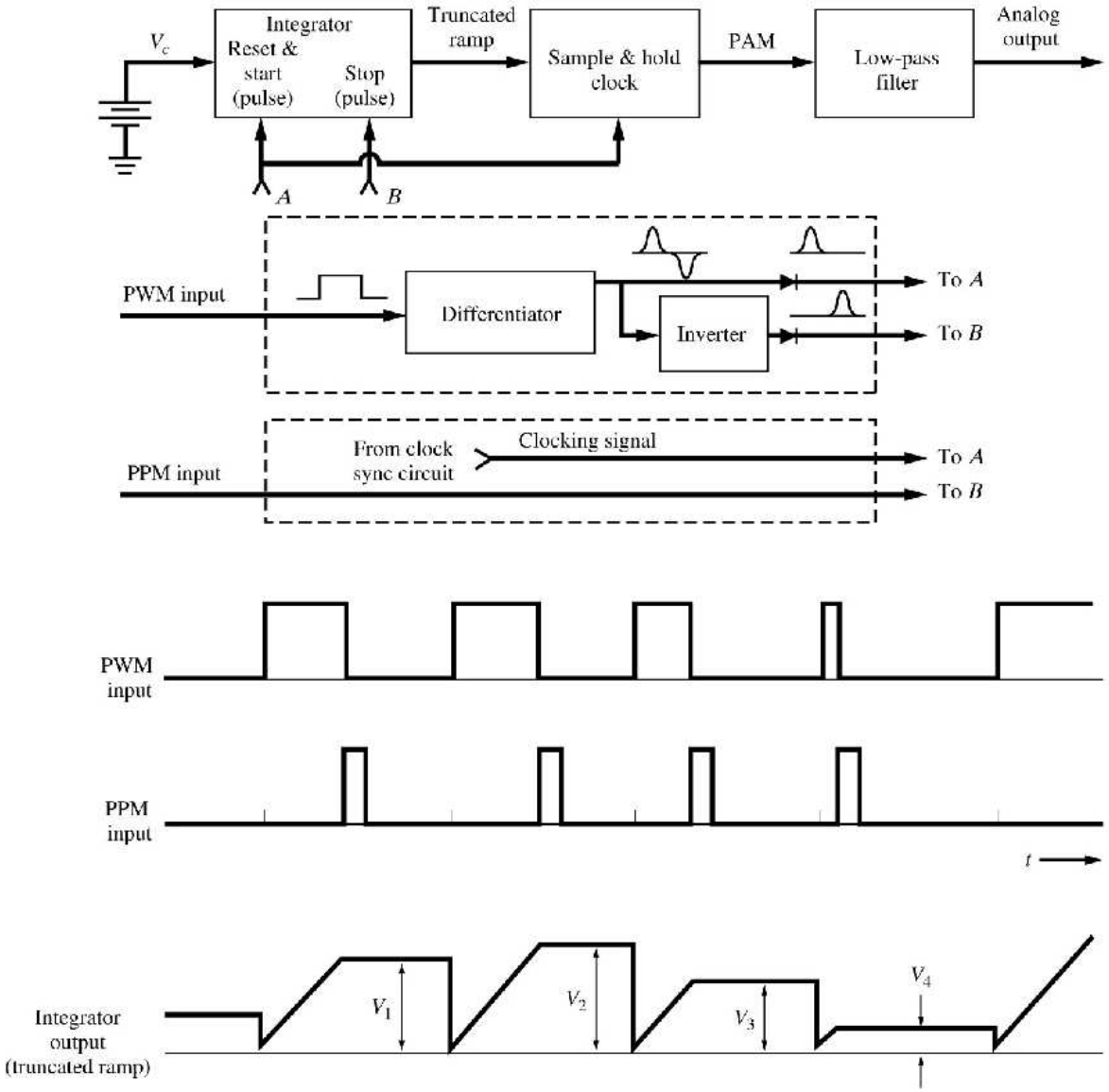
Pulse Time Modulation (4)

- The next figure shows the technique for generating naturally sampled PTM signals:



Pulse Time Modulation (5)

- The following receiving system converts PWM and PPM signals back to the corresponding analog signal:



Pulse Time Modulation (6)

- The integration starts when the PWM pulse passes from a low level to a high level.
- Integration stops when the PWM signal goes low.
- After the PWM signal goes low, the amplitude of the truncated ramp signal will be equal to the corresponding PAM sample value.
- The integrator is reset to zero at clocking times.
- In a similar way, PPM may be converted to PAM by using the clocking pulse to reset the integrator start time.
- The analog signal is generated from the PAM samples using a low-pass filter.
- PWM signals have a great immunity to additive noise compared with PAM. However, PWM signals occupy relatively large bandwidths.